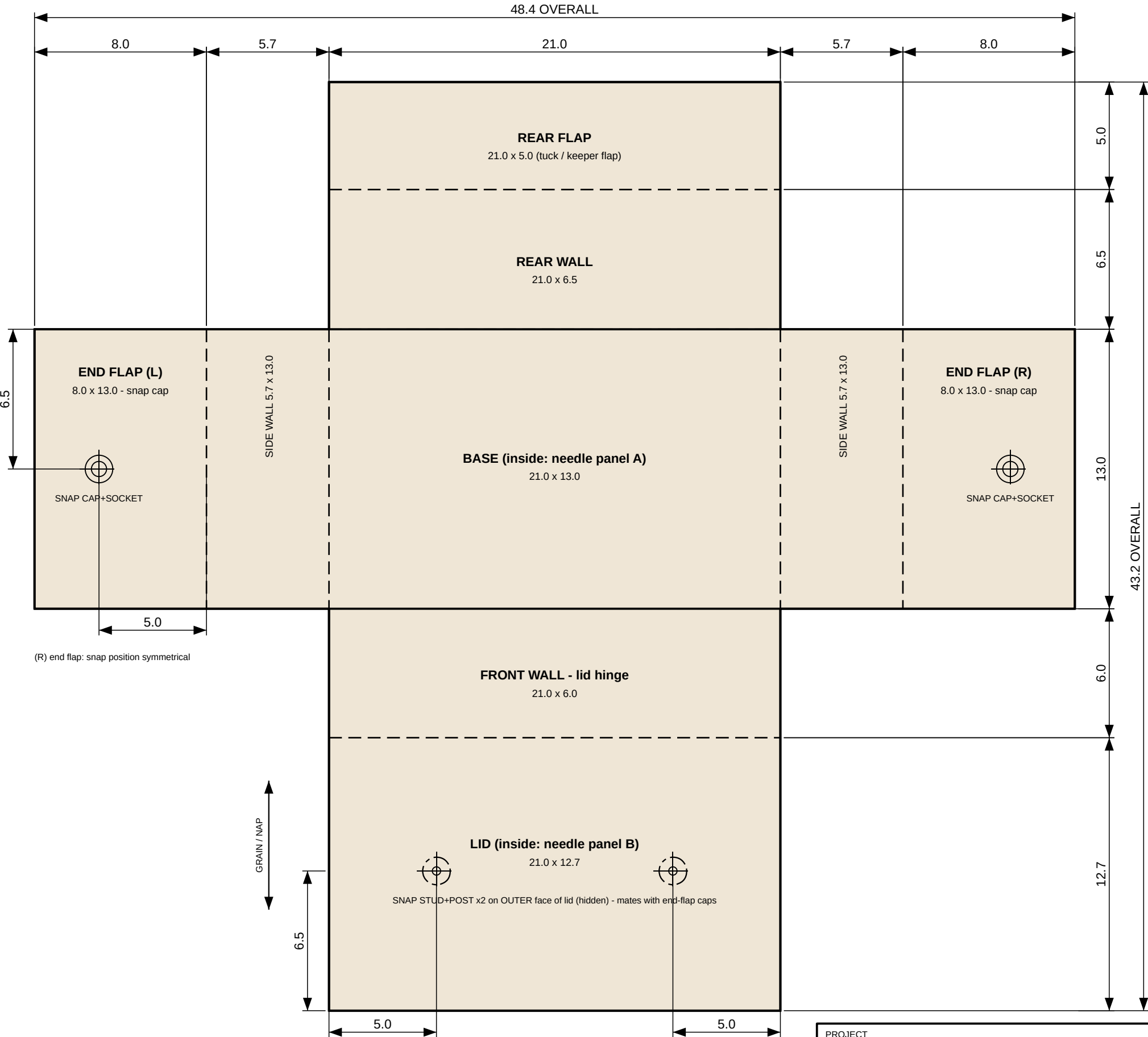


SHEET 1 - OUTER SHELL - FLAT PATTERN (DIE LINE)



LEGEND

- Cut line (net finished size)
- Fold / score line
- Hidden (feature on far face)
- Stitch line
- Snap cap + socket (this face)
- Snap stud + post (far face)

NOTES

- All dimensions in cm, finished (net) sizes as measured on the cardboard prototype. No seam allowance included: add 0.8 cm for turned seams, 0 for bound / edge-painted edges, or 1.0 cm wrap allowance if covering board.
- Pattern assumed symmetrical about its vertical centre line. Left side wall / end flap not dimensioned on the prototype; drawn equal to the right side (5.7 / 8.0).
- Wall heights are as measured: 5.7 (sides), 6.0 (front), 6.5 (rear). For a square-cornered box equalise all four to 6.0 before cutting - see sheet 2, open question Q1.
- Base 13.0 vs lid 12.7: the 0.3 cm difference lets the lid clear the rear flap when closed. Keep as measured.
- Snap: 12.5 mm (line 20) spring snaps, 2 sets on the end flaps + 2 sets on panel B keeper flap (sheet 4). Set snap studs on the lid after a dry fold of the first sample; the fold radius shifts them ~1 material thickness.
- Outer: faux leather (PU) 1.0-1.2 mm. Stiffen base, lid and walls with 1.5 mm greyboard (optional for flaps). Lining and interior panels: sheets 3-6. BOM: sheet 7.
- Rear flap (5.0) folds inward over the contents before the lid closes; end flaps fold over the lid and snap.

SHEET INDEX

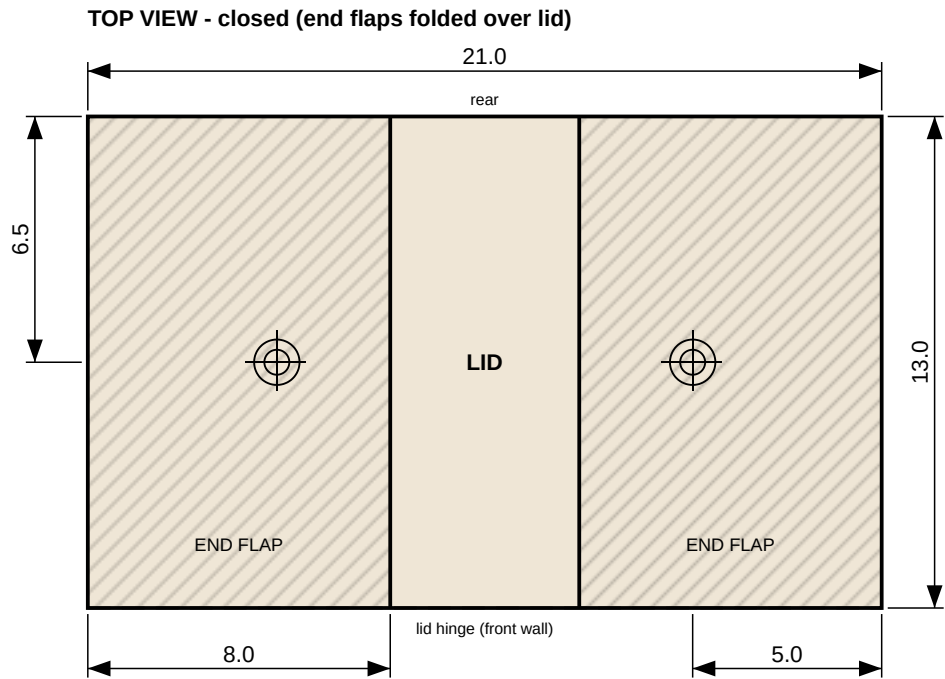
- Outer shell - flat pattern (this sheet)
- Assembly - closed views, fold sequence, interior layout
- Needle panel A (base): 10 cm tips 2.0-5.0 mm
- Needle panel B (lid): 10 cm tips 5.5-10 mm + keeper flap
- Hinged page face 1 - panel C: 5 cm tips 2.0-5.0 mm, caps, keys
- Hinged page face 2 - panel D: 5 cm tips 5.5-8 mm, accessories
- Contents checklist, bill of materials, construction

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Outer shell - flat pattern (die line)	1:2	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype, hand-measured (photos 1-5)	2026-10-01 Rev A	1 of 7

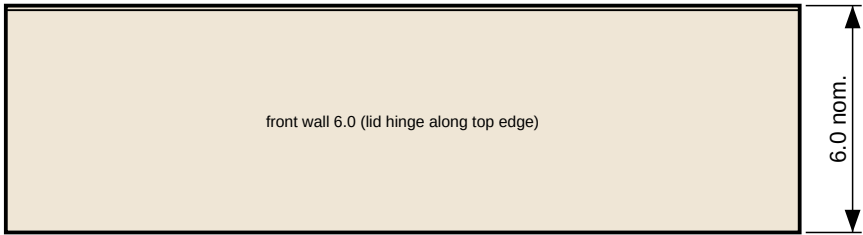
SHEET 2 - ASSEMBLY - CLOSED VIEWS, FOLD SEQUENCE, INTERIOR LAYOUT

FOLD / ASSEMBLY SEQUENCE

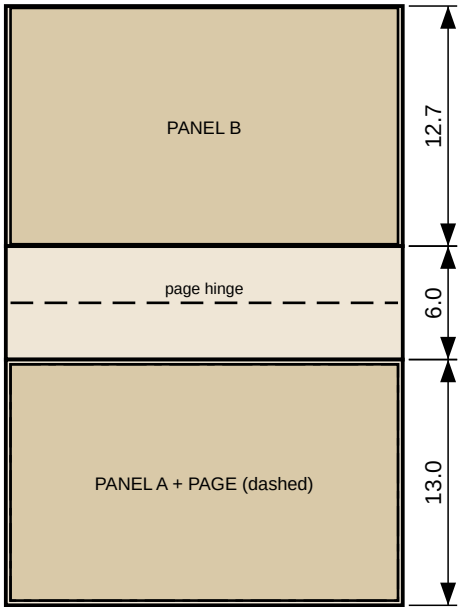
1. Fold the four walls up 90 deg from the base (rear 6.5, front 6.0, sides 5.7).
2. Glue / stitch the corner tabs (not shown; add 1.5 cm tabs on the side walls if a rigid box).
3. Fit interior: panel A to base (sheet 3), page hinge to front wall inner face (sheets 5-6), panel B to lid inner face (sheet 4).
4. Fold rear flap (5.0) inward over the contents.
5. Close lid over the front-wall hinge.
6. Fold both end flaps over the lid; snap.



FRONT VIEW - closed



INTERIOR LAYOUT - open, lid folded flat, viewed from above (1:4)

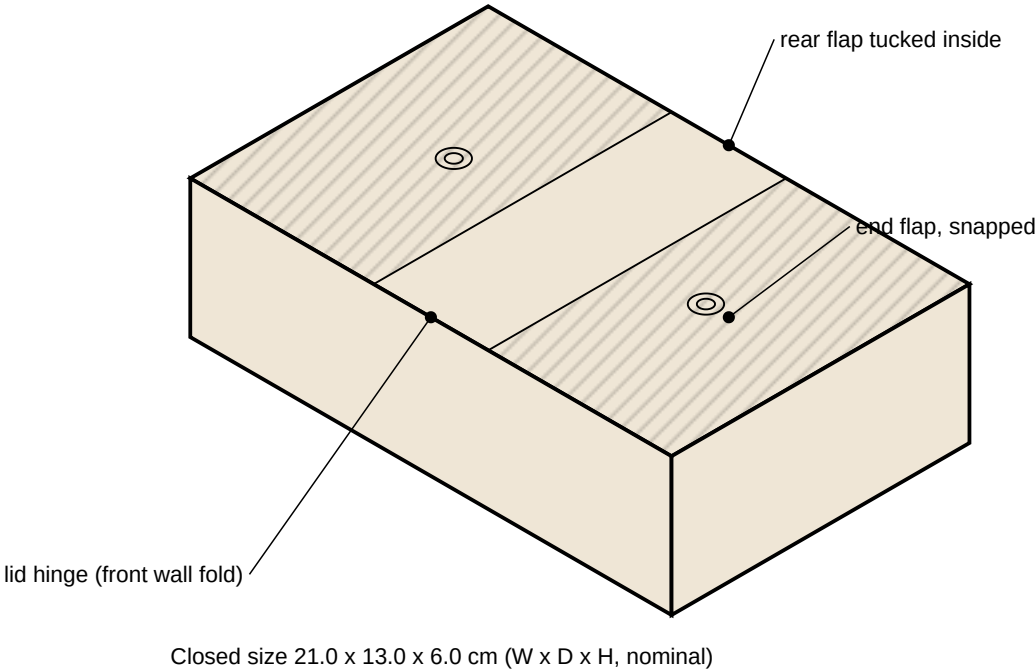


LID - panel B (sheet 4)
10 cm tips 5.5-10 mm, 6 slots, keeper flap with 2 snaps.

FRONT WALL - page hinge
1.5 cm strip; the page flips from base to lid.

BASE - panel A (sheet 3) under the hinged page
Panel A: 10 cm tips 2.0-5.0 mm, 11 slots.
Page face 1 (sheet 5): 5 cm tips 2.0-5.0 mm, caps, keys.
Page face 2 (sheet 6): 5 cm tips 5.5-8 mm, connectors, grip patch, zip cable pocket.

ISOMETRIC - closed



STACK-HEIGHT CHECK (box depth 6.0 nominal)

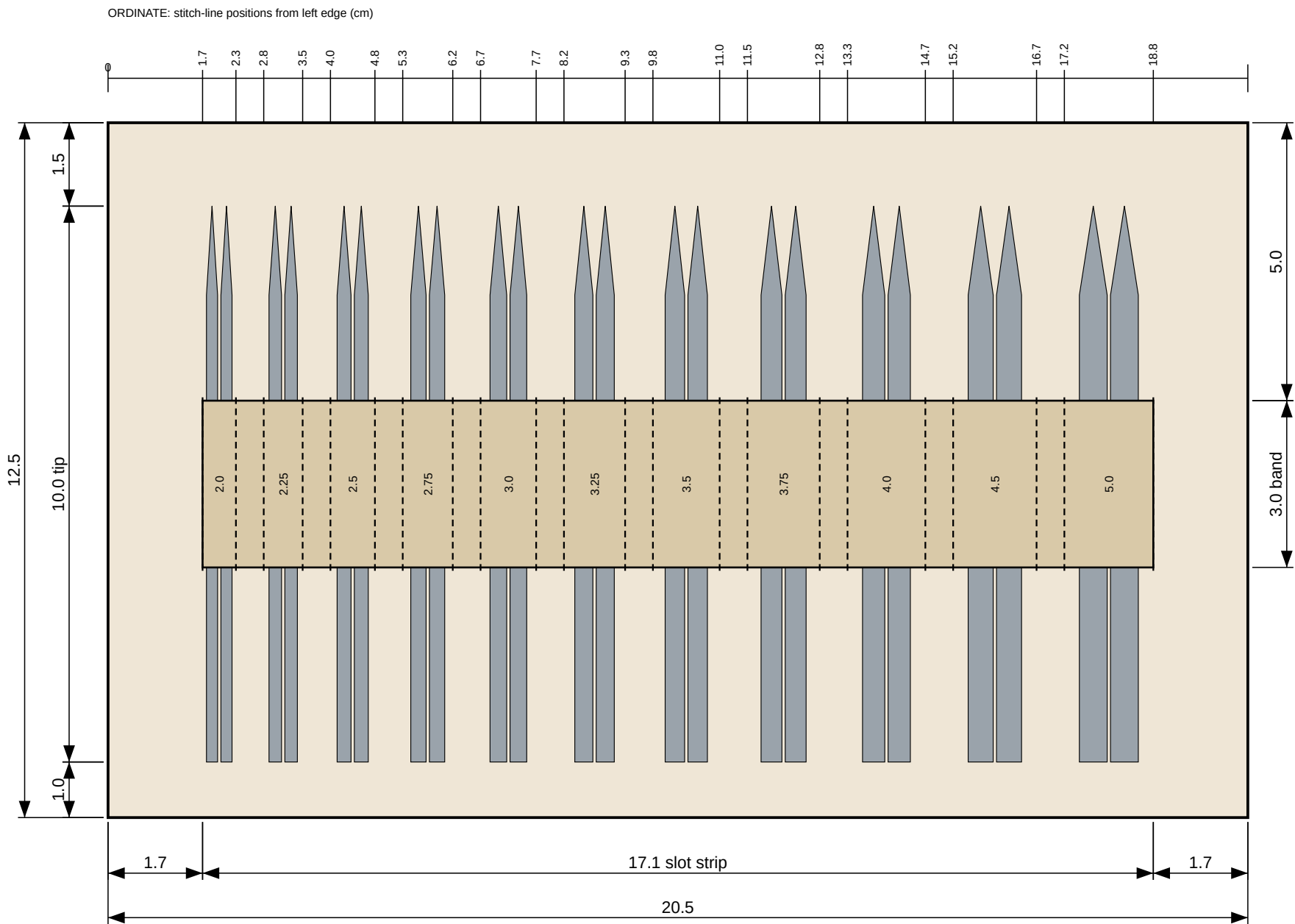
Layer (bottom to top)	Thickest item	Height
Panel A on base	2 x 5.0 mm tips side by side + band	0.8
Page face 1 (C)	2 x 5.0 mm tips + elastic	0.8
Page core	1.5 mm board + 2 lining	0.4
Page face 2 (D)	2 x 8.0 mm tips / 5 coiled cables	1.6
Panel B in lid	2 x 10 mm tips + keeper flap	1.4
Rear flap + clearance		1.0
TOTAL		6.0 (= depth, OK)

OPEN QUESTIONS (confirm before cutting)

- Q1 Keep measured wall heights 5.7 / 6.0 / 6.5 or equalise to 6.0?
Q2 Rear flap 5.0: tuck inside (drawn) or fold over the lid outside?
Q3 Soft case (faux leather + interfacing only) or board-stiffened box?
Q4 Slot bands: 25 mm knit elastic (drawn) or self faux-leather strips?
Q5 Cable pocket on the page (drawn) or a separate pouch?
Q6 Check real hardware sizes (caps, keys, connectors) against sheet 5-6 loops.

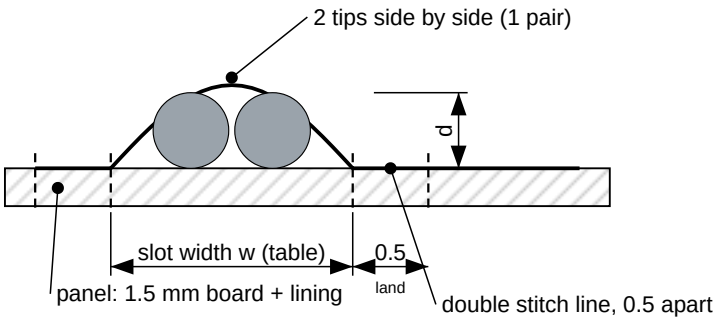
PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Assembly - closed views, fold sequence, interior layout	SCALE 1:2 (views) 1:4 (interior)	UNITS cm (page A3)
SOURCE Cardboard prototype, hand-measured (photos 1-5)	DATE / REV 2026-10-01 Rev A	SHEET 2 of 7

SHEET 3 - NEEDLE PANEL A - BASE: 10 CM TIPS 2.0-5.0 MM (11 PAIRS)



PANEL A - 20.5 x 12.5, fits BASE interior. Full size 1:1 - use as pattern.

DETAIL A - slot band section, 2:1



SLOT TABLE - panel A (positions from left edge, cm)

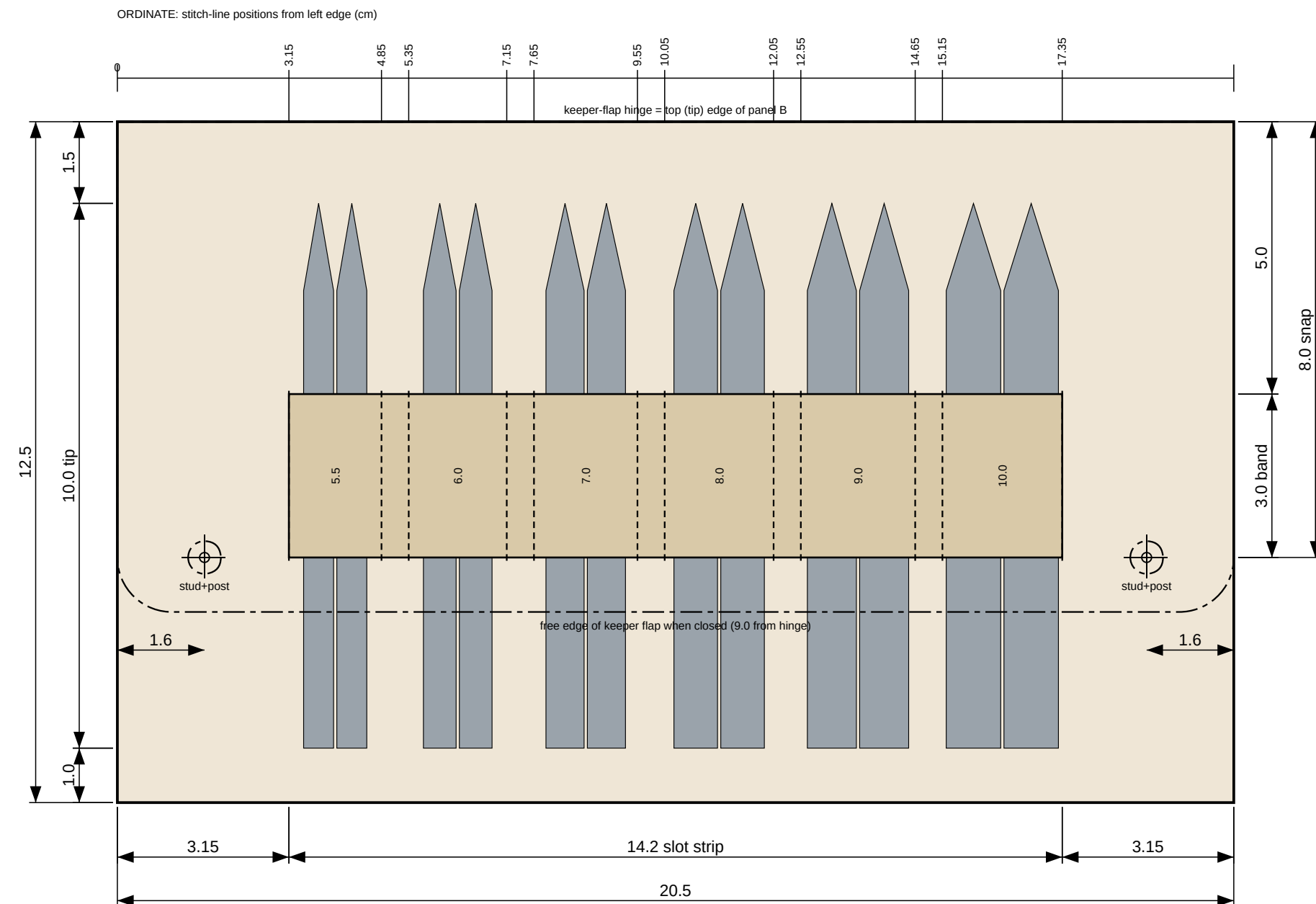
#	Size	Slot w	From	To	Holds
1	2.0 mm	0.6	1.7	2.3	2 x 10 cm
2	2.25 mm	0.7	2.8	3.5	2 x 10 cm
3	2.5 mm	0.8	4.0	4.8	2 x 10 cm
4	2.75 mm	0.9	5.3	6.2	2 x 10 cm
5	3.0 mm	1.0	6.7	7.7	2 x 10 cm
6	3.25 mm	1.1	8.2	9.3	2 x 10 cm
7	3.5 mm	1.2	9.8	11.0	2 x 10 cm
8	3.75 mm	1.3	11.5	12.8	2 x 10 cm
9	4.0 mm	1.4	13.3	14.7	2 x 10 cm
10	4.5 mm	1.5	15.2	16.7	2 x 10 cm
11	5.0 mm	1.6	17.2	18.8	2 x 10 cm

CONSTRUCTION NOTES

- Slot widths (w) are finished inside widths, taken straight from the prototype: 0.6 cm for 2.0 mm, +0.1 cm per size step, to 2.2 cm for 10 mm. Each slot holds one PAIR, the two tips side by side.
- Band: 25 mm knit elastic (drawn) stitched flat to the panel on every stitch line; no ease needed. If a faux-leather strip is used instead, add ease of 2 x tip diameter to the strip length per slot.
- Stitch lines in pairs 0.5 cm apart (the 'land'); single line at the strip ends. Stitch 3 mm, 2 passes.
- Panel = 1.5 mm greyboard wrapped in lining faux leather; finished 20.5 x 12.5 to drop into the 21.0 x 13.0 interior. Glue + edge-stitch in place.
- Size labels heat-stamped / printed on the band at each slot (shown on drawing). Order: smallest size nearest the edge shown, as on the prototype.
- Tip points toward the top edge of the drawing; 1.5 cm clear above the points (prototype).

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Needle panel A - base: 10 cm tips 2.0-5.0 mm (11 pairs)	1:1 (detail 2:1)	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype, hand-measured (photos 1-5)	2026-10-01 Rev A	3 of 7

SHEET 4 - NEEDLE PANEL B - LID: 10 CM TIPS 5.5-10 MM (6 PAIRS) + KEEPER FLAP

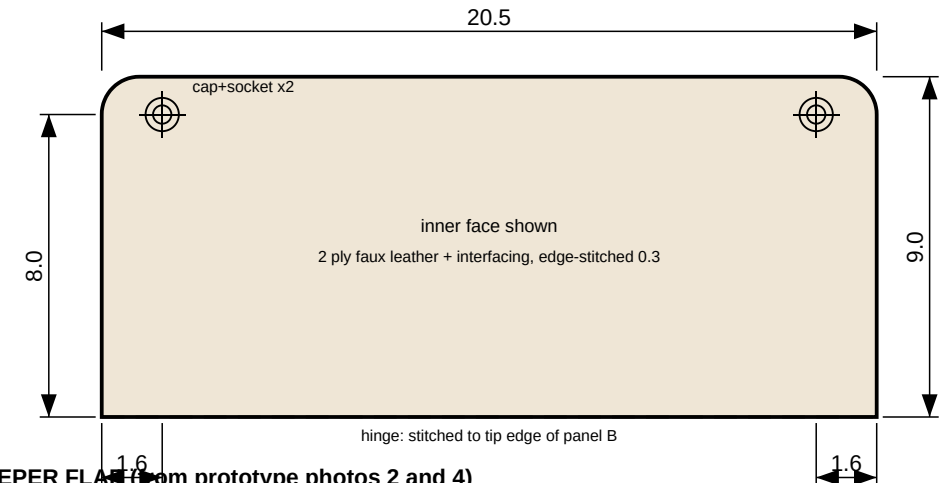


PANEL B - 20.5 x 12.5, fits LID interior. Full size 1:1 - use as pattern. Keeper flap shown folded closed (dashed) and open at 1:2, right.

SLOT TABLE - panel B (positions from left edge, cm)

#	Size	Slot w	From	To	Holds
1	5.5 mm	1.7	3.15	4.85	2 x 10 cm
2	6.0 mm	1.8	5.35	7.15	2 x 10 cm
3	7.0 mm	1.9	7.65	9.55	2 x 10 cm
4	8.0 mm	2.0	10.05	12.05	2 x 10 cm
5	9.0 mm	2.1	12.55	14.65	2 x 10 cm
6	10.0 mm	2.2	15.15	17.35	2 x 10 cm

KEEPER FLAP - open, 1:2 (20.5 x 9.0, R 1.0 free corners)



KEEPER FLAG (from prototype photos 2 and 4)

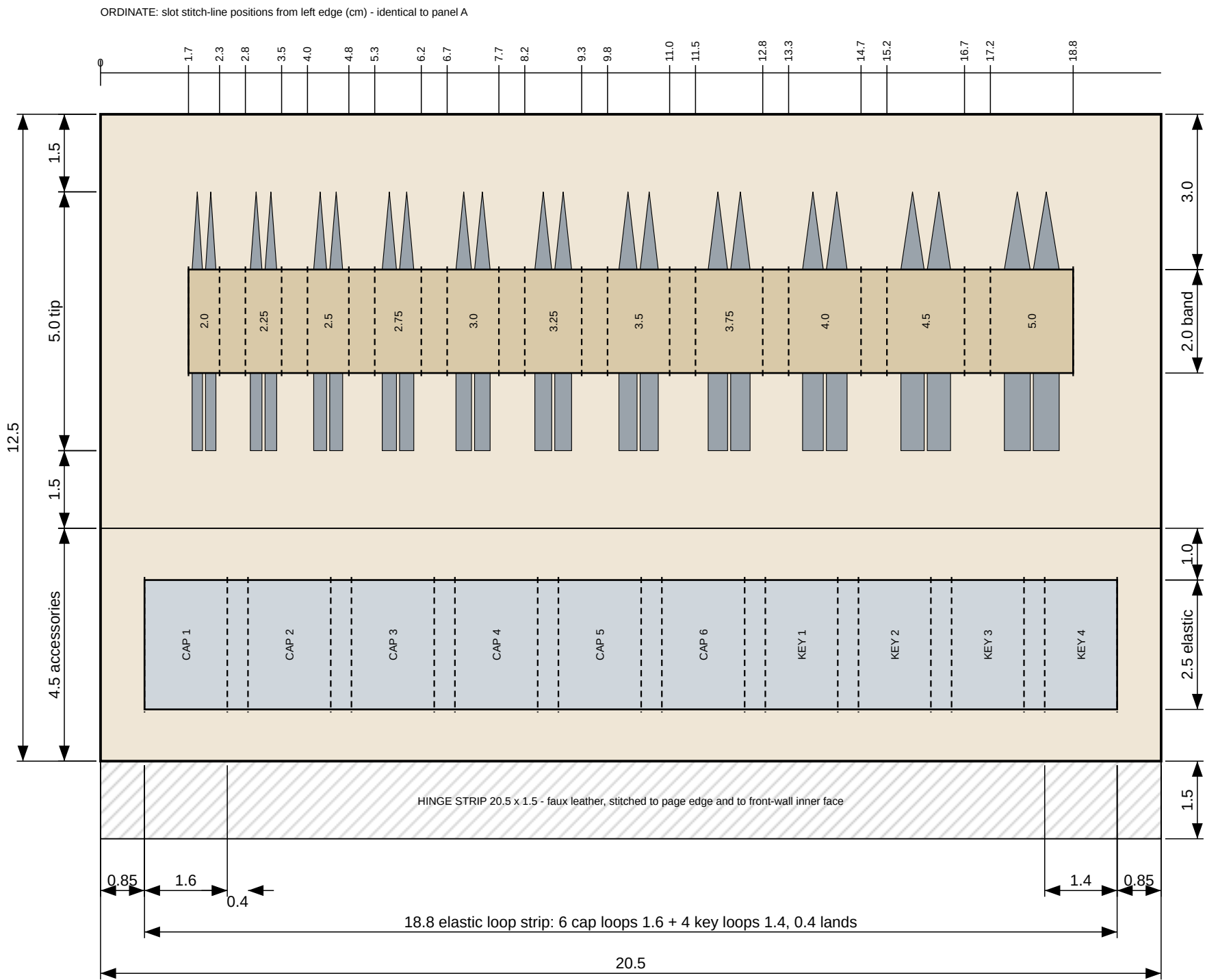
- The lid hangs open when the case is used, so the large tips need a flap: 9.0 deep (photo 2), two snaps (photo 4), hinged on the tip edge of panel B.
- Snaps sit in the 3.15 margins either side of the slot strip, 8.0 from the hinge and 1.6 from the side edges, so the studs on the panel clear the needles. Studs go through panel board + lining; back each with a 2 x 2 cm leather patch.
- Flap covers 9.0 of the 12.5 panel: the tip points (1.5-6.5 from the hinge) are fully covered.
- Check snap engagement after a dry fold; move the caps 0.1-0.2 toward the hinge if the flap pulls tight.

CONSTRUCTION NOTES

1. Slot widths (w) are finished inside widths, taken straight from the prototype: 0.6 cm for 2.0 mm, +0.1 cm per size step, to 2.2 cm for 10 mm. Each slot holds one PAIR, the two tips side by side.
2. Band: 25 mm knit elastic (drawn) stitched flat to the panel on every stitch line; no ease needed. If a faux-leather strip is used instead, add ease of 2 x tip diameter to the strip length per slot.
3. Stitch lines in pairs 0.5 cm apart (the 'land');

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Needle panel B - lid: 10 cm tips 5.5-10 mm (6 pairs) + keeper flap	SCALE 1:1 (flap 1:2)	UNITS cm (page A3)
SOURCE Cardboard prototype, hand-measured (photos 1-5)	DATE / REV 2026-10-01 Rev A	SHEET 4 of 7

SHEET 5 - HINGED PAGE, FACE 1 - PANEL C: 5 CM TIPS 2.0-5.0 MM (11 PAIRS) + CAPS & KEYS



PAGE FACE 1 - 20.5 x 12.5 (page hinge along the BOTTOM edge, 1.5 cm strip to front wall). Full size 1:1 - use as pattern.

SLOT TABLE - panel C (positions from left edge, cm)

#	Size	Slot w	From	To	Holds
1	2.0 mm	0.6	1.7	2.3	2 x 5 cm
2	2.25 mm	0.7	2.8	3.5	2 x 5 cm
3	2.5 mm	0.8	4.0	4.8	2 x 5 cm
4	2.75 mm	0.9	5.3	6.2	2 x 5 cm
5	3.0 mm	1.0	6.7	7.7	2 x 5 cm
6	3.25 mm	1.1	8.2	9.3	2 x 5 cm
7	3.5 mm	1.2	9.8	11.0	2 x 5 cm
8	3.75 mm	1.3	11.5	12.8	2 x 5 cm
9	4.0 mm	1.4	13.3	14.7	2 x 5 cm
10	4.5 mm	1.5	15.2	16.7	2 x 5 cm
11	5.0 mm	1.6	17.2	18.8	2 x 5 cm

PAGE (both faces) - proposal, not in prototype

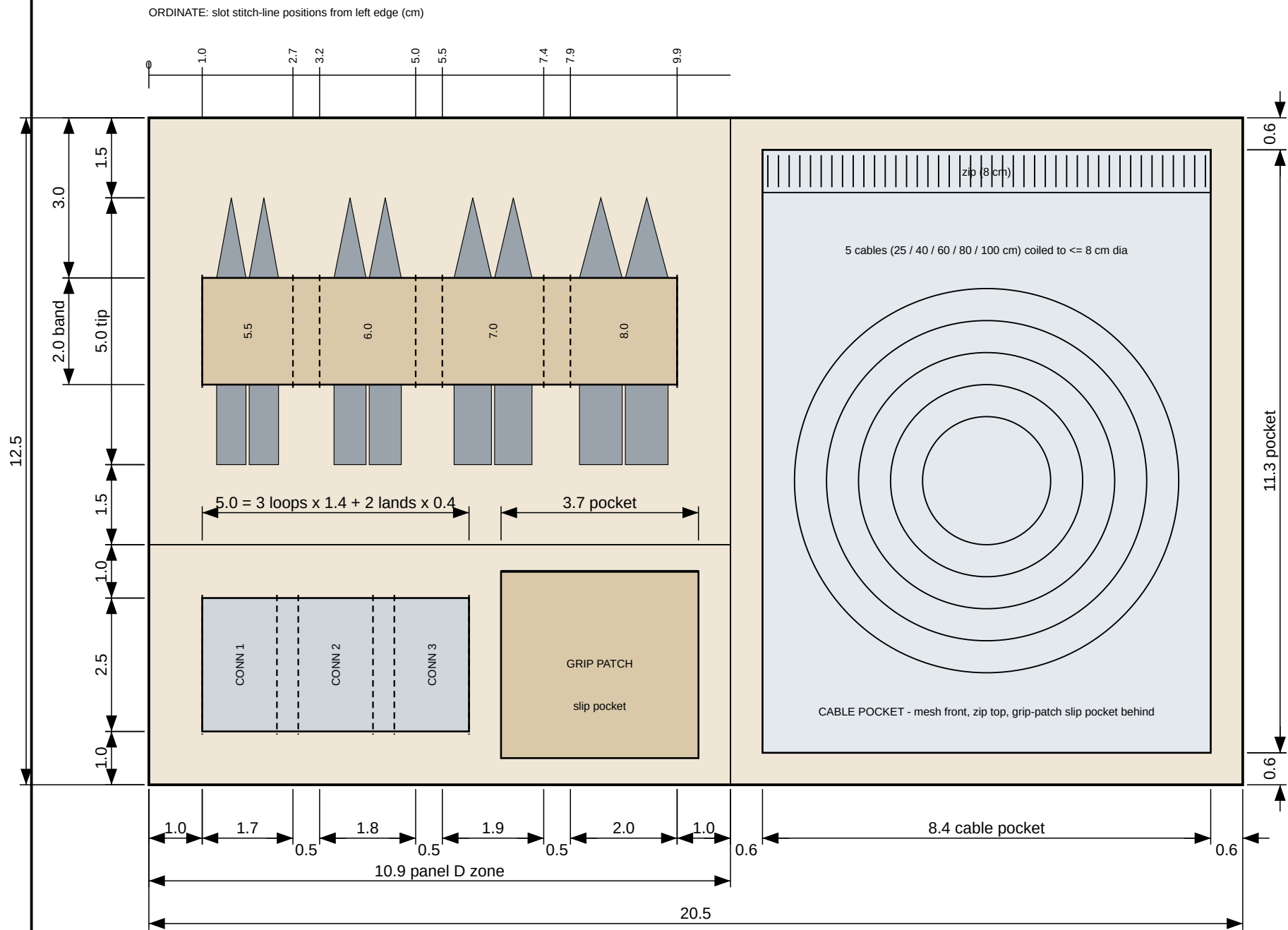
- One double-sided page carries all 5 cm tips plus the small hardware, so the 10 cm panels stay uncluttered. Core 1.5 mm board, lined both faces, 20.5 x 12.5.
- Hinge: 1.5 cm faux-leather strip along the bottom edge, stitched to the front-wall inner face. The page lies on panel A when closed and flips onto the lid when the case is opened (sheet 2).
- Face 1 (this sheet): 5 cm tips 2.0-5.0 mm in the same slot widths and positions as panel A, band 2.0 wide (narrower for the short tips). Below: 2.5 cm elastic with 6 loops for the screw-on end caps (1.6 wide) and 4 loops for the cable keys (1.4 wide).
- Loop sizes assume caps <= 1.0 dia x 1.5 long and keys <= 1.2 x 3.5. Measure the real hardware (Q6, sheet 2).

CONSTRUCTION NOTES

1. Slot widths (w) are finished inside widths, taken straight from the prototype: 0.6 cm for 2.0 mm, +0.1 cm per size step, to 2.2 cm for 10 mm. Each slot holds one PAIR, the two tips side by side.
2. Band: 25 mm knit elastic (drawn) stitched flat to the panel on every stitch line; no ease needed. If a faux-leather strip is used instead, add ease of 2 x tip diameter to the strip length per slot.

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Hinged page, face 1 - panel C: 5 cm tips 2.0-5.0 mm (11 pairs) + caps & keys	1:1	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype, hand-measured (photos 1-5)	2026-10-01 Rev A	5 of 7

SHEET 6 - HINGED PAGE, FACE 2 - PANEL D: 5 CM TIPS 5.5-8 MM (4 PAIRS) + ACCESSORIES



PAGE FACE 2 - 20.5 x 12.5 (reverse of face 1; hinge along the BOTTOM edge). Full size 1:1 - use as pattern.

SLOT TABLE - panel D (positions from left edge, cm)

#	Size	Slot w	From	To	Holds
1	5.5 mm	1.7	1.0	2.7	2 x 5 cm
2	6.0 mm	1.8	3.2	5.0	2 x 5 cm
3	7.0 mm	1.9	5.5	7.4	2 x 5 cm
4	8.0 mm	2.0	7.9	9.9	2 x 5 cm

FACE 2 LAYOUT - proposal, not in prototype

- Panel D: 5 cm tips 5.5 / 6 / 7 / 8 mm, slot widths 1.7-2.0 as on the prototype (photo 4), 1.0 margins, band 2.0 wide. Zone width 10.9.
- Below panel D: 3 connector loops (1.4 wide, 2.5 cm elastic) and a slip pocket 4.3 x 3.5 for the leather grip patch (see spec, right).
- Cable pocket 8.5 x 11.3: mesh front, 8 cm zip along the top, 0.6 cm bound edges. Holds the 5 cables coiled to <= 8 cm dia. A second slip pocket behind the mesh can take a pattern card.

LEATHER GRIP PATCH (replaces the rubber grip disc)

- Vegetable-tanned leather 1.8-2.0 mm, 6.0 x 4.0 cm, R 0.5 corners, suede side out for grip. Burnish edges.
- Optional: 0.6 cm punched hole in one corner so it can be looped onto a cable key.
- Fold it round the tip when tightening / loosening.

ELASTIC LOOP SPEC (caps, keys, connectors)

- 25 mm knit elastic stitched flat on every line; the loop widths are flat widths. Elastic stretches to take hardware up to ~1.5x the loop width.

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Hinged page, face 2 - panel D: 5 cm tips 5.5-8 mm (4 pairs) + accessories	SCALE 1:1	UNITS cm (page A3)
SOURCE Cardboard prototype, hand-measured (photos 1-5)	DATE / REV 2026-10-01 Rev A	SHEET 6 of 7

SHEET 7 - CONTENTS CHECKLIST, BILL OF MATERIALS, CONSTRUCTION & CUTTING LIST

CONTENTS CHECKLIST - 64 tips (32 pairs), 5 cables, 14 accessories: where each item lives

Size	10 cm tips	Panel	Slot	w	5 cm tips	Panel	Slot	w
2.0 mm	2 x 10 cm	A (base)	1	0.6	2 x 5 cm	C (page f.1)	1	0.6
2.25 mm	2 x 10 cm	A (base)	2	0.7	2 x 5 cm	C (page f.1)	2	0.7
2.5 mm	2 x 10 cm	A (base)	3	0.8	2 x 5 cm	C (page f.1)	3	0.8
2.75 mm	2 x 10 cm	A (base)	4	0.9	2 x 5 cm	C (page f.1)	4	0.9
3.0 mm	2 x 10 cm	A (base)	5	1.0	2 x 5 cm	C (page f.1)	5	1.0
3.25 mm	2 x 10 cm	A (base)	6	1.1	2 x 5 cm	C (page f.1)	6	1.1
3.5 mm	2 x 10 cm	A (base)	7	1.2	2 x 5 cm	C (page f.1)	7	1.2
3.75 mm	2 x 10 cm	A (base)	8	1.3	2 x 5 cm	C (page f.1)	8	1.3
4.0 mm	2 x 10 cm	A (base)	9	1.4	2 x 5 cm	C (page f.1)	9	1.4
4.5 mm	2 x 10 cm	A (base)	10	1.5	2 x 5 cm	C (page f.1)	10	1.5
5.0 mm	2 x 10 cm	A (base)	11	1.6	2 x 5 cm	C (page f.1)	11	1.6
5.5 mm	2 x 10 cm	B (lid)	1	1.7	2 x 5 cm	D (page f.2)	1	1.7
6.0 mm	2 x 10 cm	B (lid)	2	1.8	2 x 5 cm	D (page f.2)	2	1.8
7.0 mm	2 x 10 cm	B (lid)	3	1.9	2 x 5 cm	D (page f.2)	3	1.9
8.0 mm	2 x 10 cm	B (lid)	4	2.0	2 x 5 cm	D (page f.2)	4	2.0
9.0 mm	2 x 10 cm	B (lid)	5	2.1	-	-	-	-
10.0 mm	2 x 10 cm	B (lid)	6	2.2	-	-	-	-

Totals: 17 pairs x 10 cm (34 tips) on panels A + B; 15 pairs x 5 cm (30 tips) on panels C + D; 32 pairs / 64 tips.
Accessories: 6 end caps + 4 keys -> face 1 loops; 3 connectors + grip patch -> face 2; 5 cables -> face 2 pocket.

BILL OF MATERIALS (per case)

#	Item	Qty	Cut size (cm) / note
1	Outer shell, faux leather (PU) 1.0-1.2 mm	1	48.4 x 43.2 net + allowance (sheet 1)
2	Lining, faux leather 0.6-0.8 mm or cotton	1	48.4 x 43.2 net + allowance
3	Greyboard 1.5 mm: base, lid, 4 walls	6	21x13, 21x12.7, 21x6.5, 21x6, 2 of 5.7x13
4	Greyboard 1.5 mm: panels A, B, page	3	20.5 x 12.5
5	Panel lining faux leather (wrap)	3	24.5 x 16.5 (2.0 wrap)
6	Knit elastic 25 mm, slot bands	4	17.1 / 14.2 / 17.1 / 8.9 cm
7	Knit elastic 25 mm, accessory loops	2	18.8 / 5.0 cm
8	Keeper flap faux leather (2 ply)	2	20.5 x 9.0 + allowance
9	Hinge strip faux leather	1	20.5 x 3.0 (folded to 1.5)
10	Mesh, cable pocket front	1	8.5 x 11.3 + allowance
11	Zip, #3 nylon coil	1	8 cm
12	Spring snaps 12.5 mm (line 20), 4-part	4 sets	2 end flaps + 2 keeper flap
13	Snap backing patches, leather 2 x 2 cm	4	behind lid studs / panel B studs
14	Leather grip patch, veg-tan 1.8-2 mm	1	6.0 x 4.0, R 0.5
15	Thread, bonded nylon / polyester Tex 70	-	stitch length 3 mm
16	Interfacing (if soft case, no board)	1	48.4 x 43.2

CONSTRUCTION SEQUENCE

- Cut outer shell and lining from sheet 1 (net + allowance). Score fold lines on the board side.
- Make the three interior panels first (sheets 3-6): wrap board in lining, stitch bands / loops / pockets through the lining BEFORE wrapping so no stitching shows on the back. Set panel B studs.
- Set end-flap snap caps on the outer shell while flat. Studs on the lid outer face: mark from a dry fold.
- Laminate board to the outer shell (base, lid, walls); turn / bind all edges; top-stitch at 0.3 cm.
- Form the box: fold the walls, glue corner tabs or stitch the corners (box-stitch) outside-in.
- Glue + edge-stitch panel A into the base, panel B into the lid; stitch the page hinge to the front wall.
- Stitch the keeper flap to the tip edge of panel B. Check all 4 snaps engage cleanly.
- Load the set in the order of the contents checklist and check the stack height closes without pressure.

TOLERANCES

Cut pieces +/- 0.1 cm; slot stitch lines +/- 0.05 cm (the 0.6 slot must still take two 2 mm tips); snap centres +/- 0.15 cm.
Interior panels are 0.5 cm smaller than the box interior in both directions to allow for lining thickness.

PROTOTYPE MEASUREMENT RECORD - what each photo contributed

Photo	Feature	Value used on drawings
1	Outer shell cross pattern	Rear flap 5.0, rear wall 6.5, base 21.0 x 13.0, front wall 6.0, lid 12.7; side wall 5.7, end flap 8.0, snaps
2	Panel with 9 cm flap, large sizes (early draft)	Keeper flap depth 9.0; 20.5 panel length. Row widths superseded by photo 4
3	Panel 2.0-5.0 mm, 11 rows	Slot widths 0.6-1.6 (= 6-16 mm), 0.5 lands, 1.0 end margins, 1.5 side margin
4	Panel 5.5-10 mm, 6 rows + 2-snap flap	Slot widths 1.7-2.2, 0.5 lands, 0.5 top / 1.0 bottom margin, two snap tabs
5	Panel 2.0-5.0 mm, clean copy	Confirms photo 3 widths and 0.5 'between typical'; used for the 5 cm tip panel C

HARDWARE SIZE ASSUMPTIONS (loops sized to these; measure the real parts - Q6)

Item	Qty	Assumed size (cm)	Loop / pocket on drawing
Screw-on end cap	6	1.0 dia x 1.5	1.6 wide loop, face 1
Cable key	4	1.2 x 3.5	1.4 wide loop, face 1
Cable connector	3	1.0 dia x 3.0	1.4 wide loop, face 2
Leather grip patch	1	6.0 x 4.0 x 0.2	4.3 x 3.5 slip pocket, face 2 (patch folds in half)
Cables 25-100 cm	5	coil to 8.0 dia	8.4 x 11.3 zip pocket, face 2

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Contents checklist, bill of materials, construction & cutting list	SCALE -	UNITS cm (page A3)
SOURCE Cardboard prototype, hand-measured (photos 1-5)	DATE / REV 2026-10-01 Rev A	SHEET 7 of 7